Skills Summary

Pythagorean Theorem and Coordinate Distance

Use this reference while completing your worksheet or when studying.

Every problem on the worksheet is the same triangle in disguise. Find the right angle, name the longest side c, and only then decide whether you are adding squares (looking for c) or subtracting them (looking for a leg).

1. Hypotenuse from two legs (hypotenuse-from-legs)

Theorem: in a right triangle with legs a, b and hypotenuse c,

$$a^2 + b^2 = c^2 \qquad \Longrightarrow \qquad c = \sqrt{a^2 + b^2}$$

The hypotenuse is always opposite the right angle and always the longest side.

Example: A rectangle is 9 cm by 12 cm. How long is its diagonal?
Step 1. The diagonal cuts the rectangle into two right triangles whose legs are the sides, and I want the side opposite the right angle — add the squares.
Step 2. $c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$ **15** cm

Try it: Find the diagonal of an 8 cm by 15 cm rectangle.

check: **17** cm

2. A missing leg from the hypotenuse

Rearranged: $a = \sqrt{c^2 - b^2}$ — subtract when the side you want is *not* the longest one.
Ladders, ropes, guy wires and taut strings are hypotenuses: they run from a high point to a point on the ground away from the base.

Example: A 20 ft ladder rests with its base 16 ft from a wall. How high does it reach?
Step 1. The ladder is the longest side, so it is c and the height is a leg — subtract the squares.
Step 2. $h = \sqrt{20^2 - 16^2} = \sqrt{400 - 256} = \sqrt{144}$ **12** ft

Try it: A 41 ft wire runs from the top of a pole to a stake 9 ft from its base. How tall is the pole?

check: **40** ft

3. Distance between two points (coordinate-distance)

Rule: the horizontal gap $|x_2 - x_1|$ and the vertical gap $|y_2 - y_1|$ are the legs, so

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This is not a new formula — it is the theorem applied to a triangle you draw on the grid. Subtracting a negative coordinate *adds* to the gap.

Example: Distance from $(1, 2)$ to $(7, 10)$.
Step 1. Two points on a grid: build the right triangle by moving across, then up, and the distance is its hypotenuse.
Step 2. Legs $7 - 1 = 6$ and $10 - 2 = 8$, so $d = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100}$ **10** units

Try it: Find the distance from $(-3, 4)$ to $(2, -8)$.

check: **13** units

4. Multistep and composite figures (multistep-composite)

Method: break the picture into right triangles and straight runs, compute each piece, then combine at the end.

Horizontal and vertical segments need no square root. Save every rounding for the final line, and check that a shortcut is shorter than the path it replaces.

Example: Perimeter of the triangle with vertices $(0, 0)$, $(9, 0)$, $(9, 12)$.
Step 1. Two sides lie along grid lines, so only the slanted side needs the theorem; then add all three.
Step 2. Legs 9 and 12 give $\sqrt{9^2 + 12^2} = 15$, so the perimeter is $9 + 12 + 15$. **36** units

Try it: A field is 30 m by 40 m. How much shorter is the diagonal than walking along two sides?

check: **20** m

(!) Watch out: Add squares only when hunting the hypotenuse. If the answer comes out *longer* than the ladder, rope, or string in the problem, you subtracted in the wrong order. And $\sqrt{a^2 + b^2}$ is never $a + b$.